

COLEGIO DE MONTALBAN (CDM)

CDM LibHub Subsystem — Technical Features & Integration Blueprint

Subsystem Name: CDM LibHub (Mobile Subsystem)	Main System: CDM ILMS / OneServe Platform
Developer: BSIT 4C Capstone Group 9	API Backend URL: http://localhost:5002/api
App Type: React Mobile PWA / Component	Date: September 2026

1. Executive Overview & Subsystem Purpose

The **CDM LibHub Mobile Subsystem** is a specialized student-facing mobile application designed to seamlessly integrate into the CDM OneServe (Mother System) ecosystem and connect directly to the **CDM ILMS Backend**. It empowers CDM students with instant access to library services, real-time book discovery, self-service reservations, digital ID access passes, and active loan management directly from their smartphones.

2. Key Features Built & Technical Capability

- **Digital Student Access Pass & Dynamic QR Code:** Generates an ISO-18004 compliant scannable QR Code encoding student credentials for automated entry at the library kiosk scanner. Includes real-time timestamping to prevent static screenshot reuse.
- **Online Public Access Catalog (OPAC) & Search:** Enables students to search 76+ curated academic volumes across departments (ICS, IE, IBEA, CAS) with instant category filtering, availability status, and detailed book abstracts.
- **Real-Time Self-Service Book Reservations:** Allows students to submit hold requests for available volumes directly to the main library database. Instantly updates the Desktop Librarian Console for circulation approval.
- **Active Loans & Overdue Fine Tracker:** Displays active borrowed books, due dates, return statuses, and auto-computed penalty fee breakdowns to encourage timely book returns.
- **Dual-Mode Connectivity & Local Storage Cache:** Features graceful fallback to local storage mode when network connectivity is limited, ensuring student passes remain accessible offline.

3. Mother System Integration Options

The Mother System development team has **two recommended options** for embedding the Library Subsystem into the unified CDM OneServe mobile app:

Option A: Direct React Component Embedding (Recommended)

Copy the single-component source mobile-app/src/App.jsx into the Mother System project under src/modules/library/App.jsx. Import and render it inside the main tab router:

```
import LibraryMobileSubsystem from './modules/library/App.jsx';

function StudentDashboard() {
  return <LibraryMobileSubsystem studentData={currentStudentSession} />;
}
```

Option B: Independent Micro-Frontend Service

Deploy the mobile-app as an independent sub-app hosted on a dedicated port or subdomain (e.g. http://localhost:5174 or https://library.cdm.edu.ph). Embed it via web view or navigation tab inside the Mother System shell.

4. Library REST API Interface Summary

All API calls connect to the central Library Backend on port 5002 (CORS pre-configured for Mother System origins):

Endpoint	Method	Purpose	Payload / Sync
/api/books	GET	Fetch Catalog Books	Returns availability, category, ISBN, author
/api/reservations	POST	Submit Book Hold Request	Syncs instantly to Desktop Librarian Console
/api/attendance	POST	Library QR Kiosk Check-In	Logs student attendance entry timestamp
/api/students/:id/loans	GET	Get Student Borrowed Books	Returns due dates & computed penalty fees

5. Handoff Package & Delivery Checklist

- ✓ ■ **Clean Source Zip Bundle:** CDM_Mobile_App_Source_Code.zip (contains src/App.jsx, package.json, Tailwind & Vite configs).
- ✓ ■ **Integration Guide:** mobile-app/MOTHER_SYSTEM_INTEGRATION_GUIDE.md (markdown documentation).
- ✓ ■ **PDF Report Document:** CDM_Library_Mobile_App_Features.pdf (this official summary PDF).